International Journal of Infectious Diseases 163 (2026) 108238

Contents lists available at ScienceDirect

International Journal of Infectious Diseases

journal homepage: www.elsevier.com/locate/ijid

Effectiveness of the 2023-2024 monovalent XBB.1.5 vaccine against SARS-CoV-2 infection in people aged over 65 years in South Korea: a population-based cohort study

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ARTICLE INFO

Article history:
Received 8 August 2025
Revised 17 November 2025
Accepted 17 November 2025

Keywords:
COVID-19
Vaccine effectiveness
K-SEROSMART
Population-based cohort study
Nested case-control study

ABSTRACT

Objectives: The continued emergence of SARS-CoV-2 variants, including XBB.1.5 and its sublineages, poses challenges for COVID-19 control, particularly in high-risk populations. This study evaluated the community-based effectiveness of the 2023-2024 monovalent XBB.1.5 vaccine in preventing SARS-CoV-2 infection among individuals aged 65 years and older in South Korea.

Methods: This retrospective cohort study analyzed data from the K-SEROSMART, a nationwide seroepidemiological survey. The primary outcome was self-reported COVID-19 infection. To capture undiagnosed cases, this outcome was expanded to include anti-N antibody seroconversion. The follow-up period spanned from October 2023 to July 2024. Vaccine effectiveness was estimated using a Cox model with time-varying exposure for self-reported infection and conditional logistic regression for the expanded outcome based on nested case-control design with 1:4 matching.

Results: Among 3079 participants, 54.8% received the XBB.1.5 vaccine. The incidence of self-reported COVID-19 infection was lower in the vaccinated group (1.4%) than in the unvaccinated group (2.9%) ($P = 0.01$). Estimated vaccine effectiveness was 54% (95% CI: 19-74%) against self-reported infection and 33% (95% CI: 7-51%) using the expanded definition.

Conclusion: These findings support continued COVID-19 vaccination efforts, particularly in older adults. Regular vaccination remains essential to maintaining immunity and mitigating transmission, guiding future strategies against emerging SARS-CoV-2 variants.

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Introduction

The coronavirus disease 2019 (COVID-19) pandemic continues to present substantial global public health challenges, largely due to the frequent emergence of new variants of the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) virus. Identified in November 2021, the omicron variant (B.1.1.529) exhibits high trans-

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https://doi.org/10.1016/j.ijid.2025.108238
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missibility and immune escape, posing risks even to vaccinated or previously infected individuals [1]. This variant has rapidly evolved into several sublineages, with the XBB variant emerging as a dominant strain worldwide. Notably, XBB.1.5, a recombinant variant, has demonstrated increased transmissibility and immune evasion, leading to continued transmission even in highly immunized populations. As of October 2023, EG.5.1 was the most prevalent variant globally, and by late 2023, variants such as BA.2.86 and JN.1 were gaining prominence [2–4]. The ongoing antigenic drift of SARS-CoV-2 highlights the need for adaptive vaccination strategies to control transmission and severe disease.

In May and June 2023, the World Health Organization and the US Food and Drug Administration, in response to the ongoing evolution of SARS-CoV-2 omicron sublineages, recommended the development of an XBB.1.5 monovalent vaccine for the 2023/2024 season [5,6]. The 2023-2024 updated vaccines are expected to provide strong protection during the upcoming virus season, as they have demonstrated robust immune responses against the XBB lineage [7,8].

In line with these recommendations, South Korea implemented its 2023-2024 COVID-19 Vaccination Program, which began on October 19, 2023, prioritizing high-risk populations, including individuals aged 65 and older [9]. Within the first week of the rollout, over 1 million individuals in this age group had been vaccinated, accounting for 10.8% of the high-risk population—more than four times the coverage rate of the same period in the previous year (2.7%) [10]. The Korea Disease Control and Prevention Agency (KDCA) continues to emphasize the importance of vaccination in preventing severe illness and death, particularly as individuals aged 65 and older accounted for more than 30% of confirmed COVID-19 cases in October 2023. By March 2024, the XBB.1.5 vaccine coverage had reached 41.2% [11].

As XBB.1.5 continues to circulate and new subvariants (EG.5, JN.1) emerge [4], evaluating the effectiveness of the XBB.1.5 vaccine in a general population setting, particularly in high-risk groups, is essential. Assessments of the XBB.1.5 vaccine have mostly been limited to clinical settings or have focused on short-term efficacy during early vaccination stages [12–14]. Therefore, a comprehensive evaluation of its effectiveness over an extended period, including during the circulation of new variants like JN.1, is essential.

The unique characteristics of the XBB.1.5 variant, particularly its ability to evade the immune system, underscore the importance of rigorous, community-based studies to assess the effectiveness of vaccines targeting this strain. In this study, we estimated the effectiveness of the 2023-2024 monovalent XBB.1.5-targeted COVID-19 vaccine in preventing SARS-CoV-2 infection among South Korean individuals aged 65 years and older. We evaluated the long-term (7 months) effectiveness, while also considering the impact of undiagnosed infections. The findings offer critical insights for optimizing vaccination strategies in high-risk populations as new SARS-CoV-2 variants continue to emerge globally.

Methods

Data source

This retrospective cohort study utilized data from the Korea Seroprevalence Study of Monitoring of SARS-CoV-2 Antibody Retention and Transmission (K-SEROSMART), a nationwide seroepidemiological investigation conducted in South Korea. The design of the baseline study is detailed in a previous study [15]. In the present work, individuals who consented to continued participation and took part in the fifth wave of the survey, conducted in June 2024, were eligible for inclusion. The fifth-wave participants were drawn from two cohorts: one consisting of individuals whose baseline study began in the first wave (participating in the first, second, fourth, and fifth waves), and another consisting of individuals whose baseline study began in the third wave (participating in the third, fourth, and fifth waves).

The survey collected sociodemographic and health-related data, including sex, age, and underlying medical conditions. These data were supplemented with COVID-19 infection and vaccination history from the KDCA database. Because South Korea transitioned from a mandatory surveillance system to a sentinel surveillance system in August 2023, the KDCA-linked database provided confirmed cases until August 31, 2023, while subsequent cases were identified through self-reports in the survey. Serum samples were analyzed for SARS-CoV-2 antibodies using the Elecsys Anti-SARS-CoV-2 (Roche) assay, which detects spike (S) protein, and nucleocapsid (N) protein antibodies, to evaluate both vaccine- and infection-induced immune responses.

Study participants and design

The initial study population consisted of 9438 participants from the fifth K-SEROSMART study, of whom 3215 individuals aged 65 years or older were eligible for this analysis.

Exclusion criteria were as follows: (1) 60 participants whose vaccination and infection records obtained from the KDCA could not be linked due to personal identification number errors; (2) 17 cases in which the reported vaccination and infection months were identical, making it impossible to determine temporal order; (3) 39 cases reporting infection in October 2023, the start of the follow-up period, with no exposure time at risk; and (4) 6 cases with reported infections occurring at a short interval after a prior infection, which were not considered new reinfections [12]. Because infection dates collected through self-reported questionnaires were available only by calendar month, intervals between infections were therefore assessed based on month difference. After these exclusions, 3079 participants were included in the final analysis.

This study employed two complementary analytic designs: a retrospective cohort study for the primary analysis of self-reported SARS-CoV-2 infections and a nested case-control study incorporating anti-N antibody seroconversion as an expanded infection definition. For the primary cohort analysis, participants were classified according to XBB.1.5. vaccination status and followed for SARS-CoV-2 infection using self-reported survey data. The follow-up period spanned from October 2023, when the vaccination campaign began, to July 2024, at the completion of the fifth survey. Most XBB.1.5 vaccinations were administered between October and December 2023, with 98% of vaccinated participants receiving the vaccine during this period.

For the second analytic approach, an expanded infection incorporating anti-N seroconversion was applied to address the possibility of undiagnosed infections following the Korean government's downgrade of COVID-19 from a class 2 to class 4 infectious disease [16]. Because the precise timing of anti-N seroconversion was unknown, a nested case-control study was conducted to assess vaccine effectiveness under this broader outcome definition. In this analysis, cases included all participants with either self-reported infection or anti-N antibody seroconversion, while controls were selected at a 1:4 ratio from noninfected participants, matched by age (±3 years) and sex. Additionally, to explore the potential detection of reinfections among previously seropositive individuals, two sensitivity analyses were performed using alternative anti-N titer-based criteria: (1) any increase in anti-N antibody titers (1:1 matched), and (2) a $\geq$ 2-fold increase in anti-N titers (1:2 matched). Figure 1 presents the overall study design, including the exclusion criteria, cohort follow-up, and nested case-control framework, providing a comprehensive overview of the study population and analytical approach.

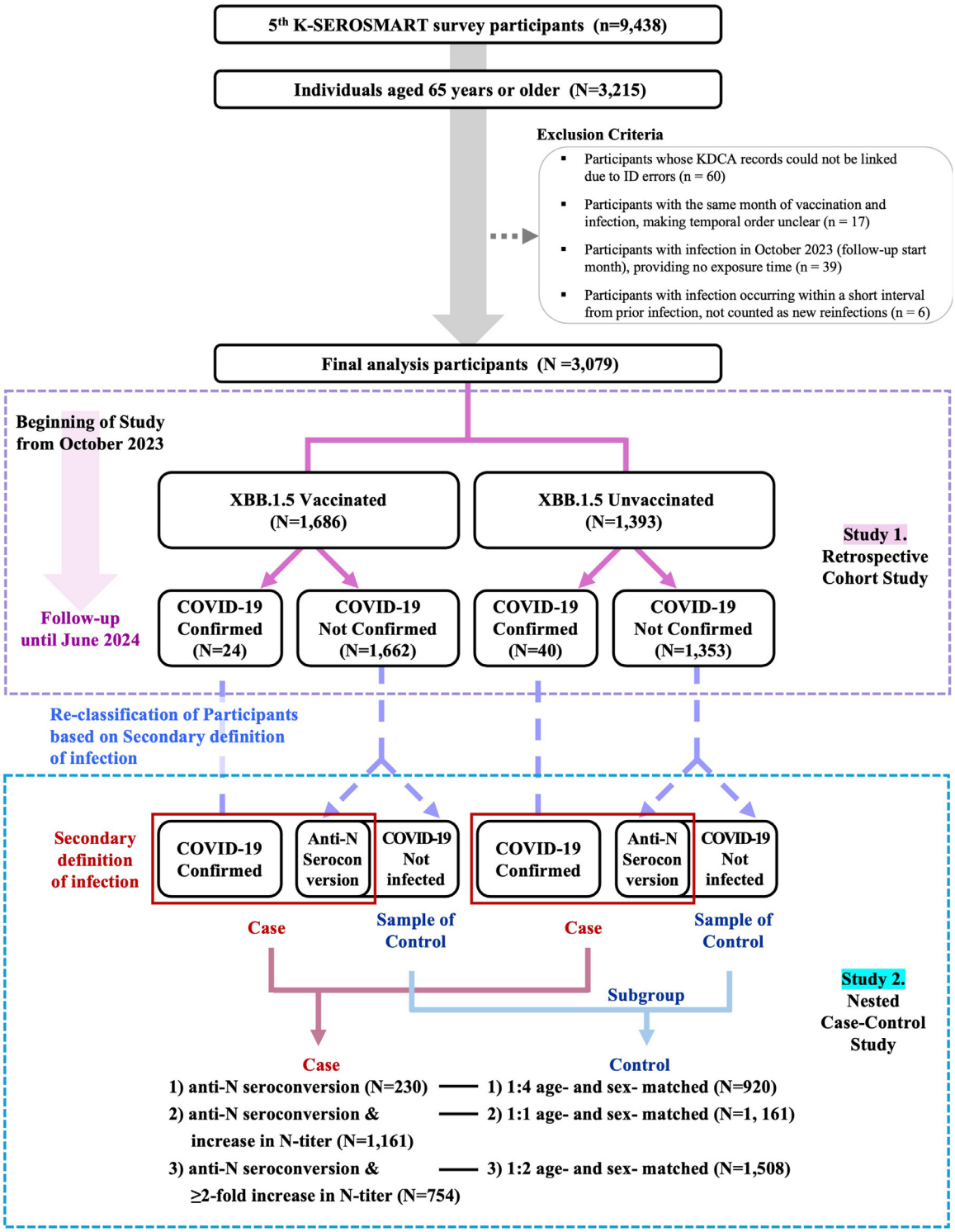

Figure 1. Flowchart of the retrospective cohort and nested case-control analyses based on participants of the 5th K-SEROSAMRT survey, October 2023-June 2024.

Outcomes

Primary definition: self-reported infection

Primary infection was based on self-reported COVID-19 cases, specifically those confirmed after October 2023, marking the start of the XBB.1.5 vaccination campaign. Participants reported the month and year of infection, specifying whether diagnosis was made by PCR or rapid antigen test (RAT). Infection data prior to December 2023 were collected in the fourth survey conducted in December 2023, and data on infections occurring thereafter were collected in the fifth survey conducted in June 2024. This was the primary outcome for evaluating vaccine effectiveness in the cohort study. Vaccine effectiveness was assessed over a 8-month period, covering the full duration from the rollout of the XBB.1.5 vaccine to survey completion.

Expanded definition: inclusion of anti-N antibody seroconversion

To complement self-reported infection and account for undiagnosed cases, an additional infection definition incorporated anti-N seroconversion seroconversion was applied. A cut-off index of

≥1.0 was used to define reactive anti-N results, indicating a positive anti-N antibody status [15]. Participants who tested nonreactive for anti-N antibodies in the fourth survey (December 2023), which was conducted during the early phase of the XBB.1.5 vaccination campaign when the vast majority of vaccinated participants (94.7%) had already received the vaccine, but reactive in the fifth survey (June 2024), were classified as having had an infection during the interval between these surveys. Thus, this seroconversion likely reflects post-vaccination infections for most participants. While this ≥ 1.0 cut-off provides a clear definition of seroconversion (new infection), there are no standardized criteria for defining reinfection based on anti-N titer increase. For exploratory analysis, two additional expanded definitions were considered: (1) participants who were already reactive in the fourth survey but showed any increase in anti-N titers in the fifth survey, and (2) participants who were already reactive in the fourth survey and showed a ≥2-fold increase in anti-N titers in the fifth survey, representing a stricter reinfection criterion. A nested case-control study design was used to assess vaccine effectiveness with this expanded definition.

Covariates

Covariates were selected based on their known or potential association with COVID-19 infection risk and vaccine effectiveness. These included age, sex, underlying medical condition, prior infection history, receipt of 2023-2024 influenza vaccine, and receipt of the bivalent vaccine, which included the Omicron-variant and ancestral strains and was recommended by the Korean government during the 2022-2023 vaccination campaign [17]. Age was categorized into two groups: 65-74 years and ≥ 75 years. Sex was analyzed as a binary variable (male or female). Underlying medical conditions were based on self-reported physician-diagnosed conditions, including hypertension, diabetes, chronic respiratory disease, cardiovascular disease, cancer, and other conditions. These variables were collected from the fifth-wave dataset and underlying medical conditions were assumed to remain unchanged during the follow-up period. Prior infection history was defined as a confirmed infection before October 2023 and was primarily obtained from the KDCA database for cases before August 2023, while infections after this date were determined through self-reported survey responses. Receipt of the bivalent vaccine, administered during the 2022-2023 vaccination campaign, was also obtained from the KDCA database. Information on receipt of the 2023-2024 seasonal influenza vaccine was also collected from the same source.

Statistical analysis

Baseline characteristics of the study population are summarized using frequencies and percentages. Differences between XBB.1.5-vaccinated and XBB.1.5-unvaccinated groups were assessed using chi-square tests or Fisher's exact tests, as appropriate. To estimate the risk of infection over time, Kaplan-Meier survival analysis was conducted, and differences in survival curves between groups were evaluated using log-rank tests. To address immortal time bias, vaccinated individuals were divided into two exposure periods: they were classified as unvaccinated before receiving the XBB.1.5 vaccine and as vaccinated after vaccination, using a time-varying exposure approach. Cox hazards regression was used to assess vaccine effectiveness when the outcome was defined as self-reported infection. Time-to-event was defined as the interval from the date of XBB.1.5 vaccination to the first confirmed infection for vaccinated cases or the survey completion date for censored participants. For unvaccinated status, the index date was set to October 2023, marking the start of the XBB.1.5 vaccination campaign, to ensure a standardized follow-up period. Time was measured in months.

For the expanded infection definition incorporating both self-reported confirmation and anti-N seroconversion, a nested case-control study was conducted. Cases included all participants with either self-reported infection or anti-N seroconversion, while controls were selected from noninfected participants, matched by age (within ± 3 years) and sex using the *MatchIt* package in R. A 1:4 matching ratio was applied when using the expanded outcome definition that included anti-N seroconversion. For the exploratory analyses incorporating reinfection definitions based on any anti-N titer increase and a ≥2-fold increase, 1:1 and 1:2 matching ratios were used, respectively. Conditional logistic regression was used to estimate the matched odds ratio (mOR) of infection. Vaccine effectiveness was calculated using the formula: vaccine effectiveness = (1 − HR [mOR]) × 100. In Cox models, hazard ratio (HR)-based estimates were adjusted for age, sex, underlying medical conditions, prior infection history, and bivalent vaccination history. Influenza vaccination status was not included as a covariate in the final model due to limited variability and strong correlation with XBB.1.5 vaccination. In conditional logistic regression, where matching variables (age and sex) were accounted for, additional adjustments were made for underlying medical conditions, prior infection history, and bivalent vaccination history. All statistical analyses were conducted using R version 4.2.1 (R Foundation, Vienna, Austria), with statistical significance defined as $P < 0.05$.

Results

General characteristics of the participants

The general characteristics of the study population differed between the vaccinated and unvaccinated groups (Table 1). Among vaccinated participants, 50.9% were female and 49.1% were male, whereas the unvaccinated group had a higher proportion of females (56.9%). A higher proportion of younger participants (Aged 65-74) was observed in the unvaccinated group (72.8% *vs.* 59.7%). Smoking status significantly differed, with more past smokers in the vaccinated group (33.5% *vs.* 25.6%) and more non-smokers in the unvaccinated group (65.5% *vs.* 59.6%). Obesity distribution and employment status were similar between the groups, with no significant differences. However, education level significantly differed, with more participants in the vaccinated group having a middle school education (57.8%) and more in the unvaccinated group having a college degree. Underlying medical conditions were more common in the vaccinated group (69.1% *vs.* 61.2%). Influenza vaccination was more common in the vaccinated group (98.0%) compared to the unvaccinated group (82.8%), and bivalent vaccination rates were also significantly higher among the vaccinated (67.1% *vs.* 28.6%). Following October 2023, the incidence of self-reported infection was significantly lower in the vaccinated group than in the unvaccinated group (1.4% *vs.* 2.9%, $P = 0.007$). When applying the expanded outcome definition that included both self-reported infection and anti-N antibody seroconversion, the infection rate remained significantly lower in the vaccinated group (6.2% *vs.* 9.0%, $P = 0.005$).

Figure 2 presents Kaplan-Meier survival curves comparing the risk of COVID-19 infection between individuals who ever received the XBB.1.5 vaccine and those who remained unvaccinated, based on static exposure classification. Although this approach does not account for time-varying vaccination status as in the main analysis, it provides a descriptive comparison of cumulative incidence between groups. Vaccinated individuals showed a lower probability of infection compared to unvaccinated individuals (log-rank test, $P = 0.006$). Follow-up time was expressed as months since October, 2023 for unvaccinated individuals and since the date of XBB.1.5

Table 1
General characteristics of study participants aged 65 years and older from the fifth K-SEROSMART study by XBB.1.5 vaccination status, October 2023-June 2024.

Variable	XBB vaccinated (N = 1686, 54.8%)	XBB not vaccinated (N = 1393, 45.2%)	*P*-value
Sex			<0.001
Male	828 (49.1)	600 (43.1)	
Female	858 (50.9)	793 (56.9)	
Age group			<0.001
Aged 65-74	1007 (59.7)	1014 (72.8)	
Aged >75	679 (40.3)	379 (27.2)	
Smoking status			<0.001
Non-smoker	1005 (59.6)	912 (65.5)	
Past smoker	565 (33.5)	357 (25.6)	
Current smoker	116 (6.9)	124 (8.9)	
Obesity status			0.247
Underweight	49 (2.9)	34 (2.4)	
Normal weight	602 (35.7)	533 (38.3)	
Overweight	527 (31.3)	396 (28.4)	
Obese	508 (30.1)	430 (30.9)	
Employment status			0.140
Employed	560 (33.2)	499 (35.8)	
Unemployed	1126 (66.8)	894 (64.2)	
Education level			<0.001
Middle school	975 (57.8)	685 (49.2)	
High school	500 (29.7)	453 (32.5)	
College/University	211 (12.5)	255 (18.3)	
Underlying medical condition			<0.001
No	521 (30.9)	541 (38.8)	
Yes	1165 (69.1)	852 (61.2)	
Influenza vaccination			<0.001
No	34 (2.0)	239 (17.2)	
Yes	1652 (98.0)	1153 (82.8)	
Bivalent vaccination			<0.001
Not vaccinated	554 (32.9)	995 (71.4)	
Vaccinated	1132 (67.1)	398 (28.6)	
Prior infection status (pre-Oct 2023)			0.829
No	730 (43.3)	603 (43.3)	
Yes	956 (56.7)	790 (56.7)	
Infected after Oct 2023 (self-reported PCR or RAT)			0.007
No	1662 (98.6)	1353 (97.1)	
Yes	24 (1.4)	40 (2.9)	
Infected after Oct 2023 (self-reported or anti-N seroconversion)			0.005
No	1581 (93.8)	1268 (91.0)	
Yes	105 (6.2)	125 (9.0)	

Table 2
XBB.1.5 vaccine effectiveness of the monovalent against self-reported SARS-CoV-2 infection in individuals aged ≥65 years: cohort analysis from the 5th K-SEROSMART, October 2023-June 2024.

	Number of cohort at baseline	Person-month	Number of events	Incidence rate (per 100 Person-month)	Age- and sex-adjusted HRs[1]	Fully adjusted HRs[2]	Vaccine effectiveness[3]
XBB vaccinated	1686	11,725	40	0.35 (0.25-0.47)	0.56 (0.34-0.92)	0.47 (0.27-0.83)	53 (17 to 73)
Not vaccinated	1393	11,551	24	0.20 (0.13-0.30)	REF	REF	REF

[1] Models adjusted for age group and sex.
[2] Models adjusted for age group, sex, underlying medical conditions, prior infection history, and bivalent vaccination history (from the prior season).
[3] Vaccine effectiveness was calculated using fully adjusted HRs.

vaccination for vaccinated individuals. Because vaccination timing varied, follow-up was generally shorter in the vaccinated group, leading to a faster drop in the number at risk.

Effectiveness of the XBB.1.5 monovalent vaccine

Table 2 presents the overall effectiveness of the XBB.1.5 vaccine against self-reported COVID-19 infection. For the outcome defined by self-reported infection, the vaccine demonstrated an effectiveness of 53% (95% confidence interval [CI]: 17-73%) in preventing infection.

Table 3 presents the odds ratios and vaccine effectiveness estimates across three outcome definitions that incorporated anti-N antibody serologic data alongside self-reported infection. For the primary expanded definition using anti-N seroconversion (*i.e.*, change from nonreactive to reactive), 229 cases and 916 controls were included in a 1:4 age- and sex-matched nested case-control analysis. The general characteristics of this matched subset are shown in Supplementary Table 1. To further explore potential reinfections, two additional sensitivity analyses were conducted using anti-N antibody titers: (1) any increase in anti-N titers among previously seropositive individuals and (2) a ≥2-fold increase. These analyses included 2322 participants (1161 cases and 1161 controls, 1:1 matched) and 2262 participants (754 cases and 1508 controls, 1:2 matched), respectively. The general characteristics of these matched subsets are presented in Supplementary Tables 2 and 3.

The estimated vaccine effectiveness was 33% (95% CI: 7-51%) for anti-N seroconversion alone, 27% (95% CI: 13-39%) for the broader reinfection definition using any titer increase, and 23%

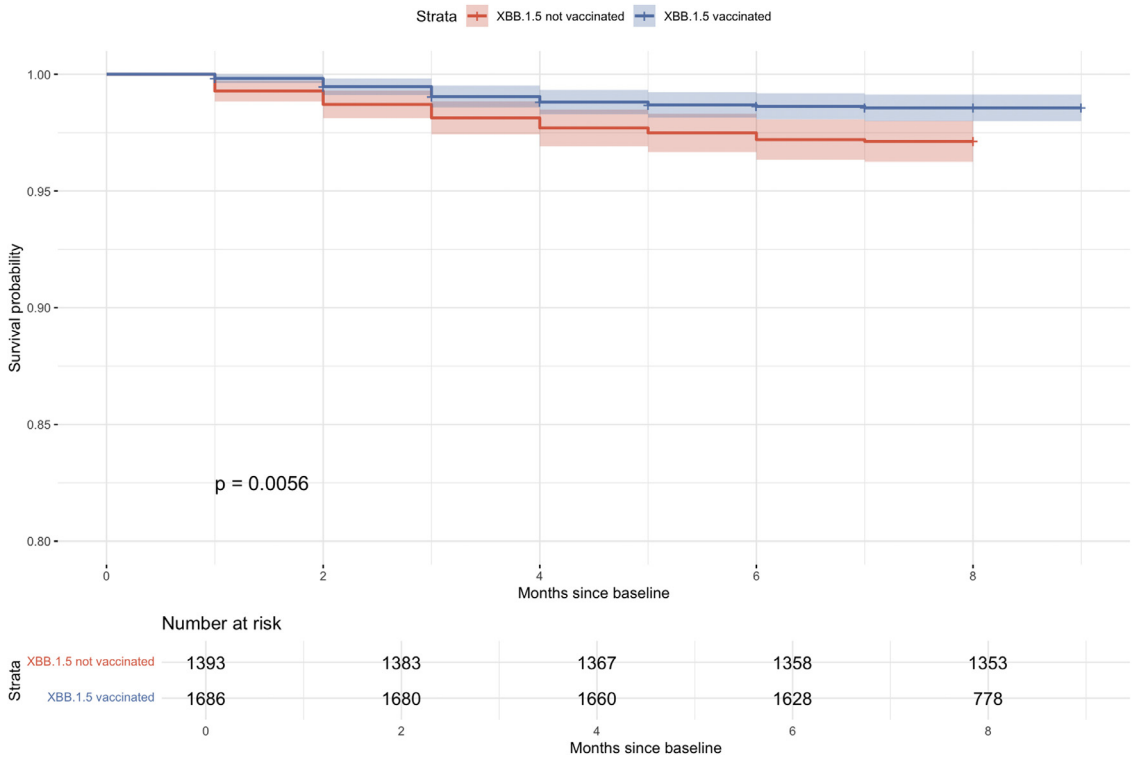

Figure 2. Kaplan-Meier survival curves comparing COVID-19 infection risk by XBB.1.5 vaccination status.
This figure does not account for changes in vaccination status over time; vaccinated individuals are classified based on ever receiving the XBB.1.5 vaccine. The analysis is for descriptive purposes only and differs from the time-varying Cox model used in the main analysis. Follow-up time is presented in months since October 19, 2023 (for unvaccinated individuals) or since vaccination (for vaccinated individuals).

Table 3
XBB.1.5 vaccine effectiveness of the monovalent against self-reported or serologically defined SARS-CoV-2 infection in individuals aged ≥65 years: nested-case control analysis from the 5th K-SEROSMART, October 2023-June 2024.

	Cases, n (%)	Controls, n (%)	Crude mORs[4]	Fully adjusted mORs[5]	Vaccine effectiveness[6]
	Case definition 1: Self-reported cases or Anti-N antibody seroconversion[1]				
XBB vaccinated	105 (45.7)	493 (53.6)	0.73 (0.54-0.97)	0.67 (0.49-0.93)	33% (7 to 51)
Not vaccinated	125 (54.3)	427 (46.4)	REF	REF	REF
	Case definition 2: Self-reported cases or Anti-N antibody seroconversion or Any anti-N titer increase[2]				
XBB vaccinated	568 (48.9)	654 (56.3)	0.73 (0.62-0.87)	0.73 (0.61-0.87)	27% (13 to 39)
Not vaccinated	593 (51.1)	507 (43.7)	REF	REF	REF
	Case definition 3: Self-reported cases or Anti-N antibody seroconversion or ≥2-fold anti-N titer increase[3]				
XBB vaccinated	367 (48.7)	822 (54.5)	0.79 (0.66-0.94)	0.77 (0.63-0.93)	23% (7 to 37)
Not vaccinated	387 (51.3)	686 (45.5)	REF	REF	REF

[1] Anti-N antibody seroconversion was defined as a change from nonreactive (<1.0 cut-off index) in the fourth survey to reactive (≥1.0) in the fifth survey.
[2] Among participants who were already reactive in the fourth survey, any increase in anti-N antibody titers measured in the fifth survey.
[3] Among participants who were already reactive in the fourth survey, a ≥2-fold increase in anti-N antibody titers compared to the fourth survey, representing a stricter criterion for suspected reinfection.
[4] Age and sex were matched prior to analysis and thus not further adjusted in the models.
[5] Models were adjusted for underlying medical conditions, prior infection history, and bivalent vaccination.
[6] Vaccine effectiveness was calculated using fully adjusted mORs.

(95% CI: 7-37%) when applying the ≥2-fold increase criterion. The XBB.1.5 vaccine continued to show a protective effect, even under broader outcome definitions that incorporated serological evidence, although the estimated effectiveness was lower than that based on self-reported infections alone.

Discussion

In this population-based cohort study, we evaluated the effectiveness of the 2023-2024 monovalent XBB.1.5 vaccine in preventing SARS-CoV-2 infection among individuals aged 65 years and older in South Korea. The vaccine demonstrated stable effectiveness at 53% when infection was defined solely by self-reported cases, based on a time-varying Cox model that adjusted for immortal time bias and thus reinforced the robustness of this estimate. To better capture total infection burden, including undiagnosed cases, anti-N antibody seroconversion was also incorporated into the outcome definition. With this broader definition, vaccine effectiveness declined to 33%, but remained statistically significant.

Unlike prior studies that have had shorter follow-up periods, this study provides a comprehensive evaluation over the full duration of the XBB.1.5 vaccination campaign in South Korea. Previous hospital-based, test-negative, case-control studies have reported vaccine effectiveness rates of 56.8% across all ages over

1 month [13] and 60.2% in individuals aged 65 and older over 2 months [18]. These higher estimates likely reflect the focus of these studies on more severe cases, where vaccine protection is stronger, whereas our community-based cohort included a broader spectrum of infections, including milder cases. Additionally, previous studies have assessed vaccine effectiveness over a shorter period, when vaccine-induced immunity is at its peak. In contrast, we followed participants for 7 months, allowing for a more comprehensive assessment of effectiveness over time. Despite these differences, our findings align closely with international studies, which have reported vaccine effectiveness rates of 46% in the United States among individuals aged 50 and older [12] and 50% in the Netherlands among those aged 60 and older [19]. This consistency across different populations and study designs strengthens the reliability and generalizability of the vaccine's protective effects.

To provide a more inclusive assessment of vaccine effectiveness, infection was defined using both self-reported cases and anti-N antibody conversion, allowing detection of undiagnosed infections that may have been asymptomatic or missed due to changes in surveillance policies. This approach enabled a more comprehensive evaluation of vaccine effectiveness in a real-world, community-based setting. While the expanded infection definition resulted in lower observed vaccine effectiveness, the XBB.1.5 vaccine still provided significant protection. This aligns with prior evidence that COVID-19 vaccines provide stronger protection against symptomatic or severe disease than against asymptomatic or mild infections [20]. The inclusion of mild or asymptomatic cases—more likely to be captured through serological testing—may explain the reduced effectiveness observed. Moreover, vaccinated individuals who become infected may have experienced attenuated symptoms due to vaccine-induced immunity [21,22], making them less likely to report or be diagnosed with infection. Finally, some infections identified through antibody seroconversion may have occurred at relatively later time points after vaccination, when vaccine-induced protection against infection tends to decline [20], further contributing to the lower observed effectiveness.

In addition, we conducted sensitivity analyses using alternative serological definitions of potential reinfection, based on significant increases in anti-N antibody titers among individuals who were already seropositive at baseline. These analyses identified more reinfection-like cases, but the vaccine remained consistently protective across definitions, reinforcing the robustness of our findings. However, in the absence of standardized criteria for serology-based reinfection, these results should be interpreted with caution, and further work is needed to validated serological thresholds and their epidemiologic relevance in vaccine effectiveness studies.

The sustained protective effect observed in this study reinforces the importance of regular COVID-19 vaccinations to maintain immune protection, particularly in older adults. Vaccine effectiveness was maintained over the 8-month follow-up, indicating durable immunity during this period. With the ongoing emergence of new variants and potential seasonal resurgence of SARS-CoV-2, maintaining immunity through periodic booster vaccinations remains crucial for protecting high-risk populations. Aligning vaccination campaigns with seasonal epidemic trends and observed patterns of immune response may enhance their effectiveness, particularly in high-risk populations [23]. Additionally, sustained vaccination efforts provide economic benefits by reducing healthcare costs associated with hospitalization and severe outcomes [24]. Public health strategies should adapt to evolving epidemiological patterns to ensure long-term immunity and preparedness against future outbreaks. Future research should focus on determining the optimal booster intervals and evaluating vaccine effectiveness against emerging subvariants to guide vaccination policies in aging populations.

This study had several strengths. First, the population-based design enhances generalizability to the broader older adult population in South Korea. Second, our results provide a community-based assessment of vaccine effectiveness by integrating both self-reported and serologically confirmed infections, offering a more comprehensive evaluation of vaccine-conferred protection. However, some limitations should also be acknowledged. First, we focused on vaccine effectiveness against infection, rather than its impact on severe disease outcomes. Given the strong association between SARS-CoV-2 infection and COVID-19-related mortality in older adults, these findings still provide valuable indirect insights into the potential role of the vaccine in reducing severe cases. Second, although the exact timing of infection could not be determined for cases identified by anti-N seroconversion, the potential for temporal misclassification is likely limited because the vast majority vaccinated participants received the XBB.1.5 vaccine before the baseline survey. Thus, seroconversion observed thereafter likely reflects post-vaccination infections. Moreover, the expanded outcome definition increased infection counts more in the unvaccinated group, which would bias estimates toward the null rather than inflate vaccine effectiveness. Third, we did not distinguish between specific SARS-CoV-2 sublineages, meaning that variant-specific effectiveness remains uncertain. Although we conducted an exploratory, time period-based analysis informed by publicly available variant surveillance trends from the KDCA, our dataset lacked individual-level variant information, and most vaccination (98%) occurred during October—December 2023 when XBB sublineages (*e.g.*, HK.3) predominated. The small number of infections during HK.3 period and the potential for misclassification limited the interpretability of the results. Therefore, these findings were not included in the main results. Given recent evidence that XBB.1.5 vaccines have lower—but still present—effectiveness against emerging variants such as JN.1 compared to earlier sublineages [8], this remains an important area for future research. Despite these limitations, our findings offer critical evidence for optimizing vaccination strategies in high-risk populations.

Conclusions

Our findings highlight the community-based effectiveness of the 2023-2024 monovalent XBB.1.5 vaccine in reducing SARS-CoV-2 infection risk among older adults in South Korea. Despite variation in effectiveness based on the infection definition employed, the vaccine consistently provided substantial protection. These findings underscore the importance of timely and routine booster vaccinations, particularly for individuals relying on vaccine-induced immunity. To enhance protection in aging populations as new variants emerge, vaccination strategies should focus on community-level outcomes. Serological assessments can complement surveillance by detecting unreported infections and helping to evaluate the full spectrum of vaccine impact, thereby informing public health strategies to protect high-risk groups.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research was supported by the National Institutes of Health of the Korea Disease Control and Prevention Agency (grant no. 2023E260500). The funder had no role in the study design; data

collection, analysis, or interpretation; manuscript preparation; or the decision to submit the manuscript for publication.

Ethical approval

This study received an exemption from Institutional Review Board review by the KDCA, in accordance with Article 36 of the Bioethics Act, as it was deemed an urgent public health study conducted by or commissioned by the local government (KDCA-2023-11-01-R-02).

Author contributions

Conceptualization: SJ, DHK. Data curation: SJ, SL, HK, HW, DHK. Formal analysis: SJ. Investigation: SJ, SL, HK, HW, DHK. Methodology: SJ, DHK. Project administration & Funding acquisition: DHK. Supervision: DHK. Writing—original draft: SJ, HK. Writing—review & editing: SJ, GR, HSJ, JWL, ARK, BCC, KRP, DHK. All authors had full access to the data, reviewed and edited the manuscript, and accept responsibility for submitting the paper for publication.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijid.2025.108238.

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